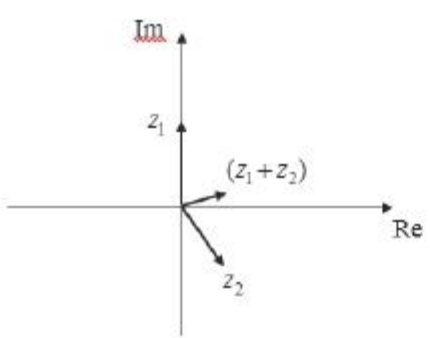


Given: Due:	Further Maths (Core Pure) HW 02	Name:
Chapter 1 & 2: Complex Numbers: modulus-argument		
1)		
$z_1 = 3i \text{ and } z_2 = \frac{6}{1+i\sqrt{3}}$		
(a) Express z_2 in the form $a + ib$, where a and b are real numbers.		(2)
(b) Find the modulus and the argument of z_2 , giving the argument in radians in terms of π .		(4)
(c) Show the three points representing z_1 , z_2 and $(z_1 + z_2)$ respectively, on a single Argand diagram.		(2)

Almost all students made a confident attempt at part (a). However, in part (b) a significant number of students neglected to calculate the modulus, indicating that they did not read the question properly. When the modulus was used it was usually found correctly. Almost all students recognised the use of the tan function to calculate the argument. However, many students did not make use of an argand diagram or their knowledge of complex numbers to use the correct sign for the argument, leaving their answer as $\pi/3$. The Argand diagram in part (c) was frequently completed successfully, although not all students could place z_2 in the correct quadrant, even if they had correctly calculated part (a).

Question Number	Scheme	Marks
(a)	$z_2 = \frac{6(1-i\sqrt{3})}{(1+i\sqrt{3})(1-i\sqrt{3})} = \frac{6(1-i\sqrt{3})}{4}$ $z_2 = \frac{6(1-i\sqrt{3})}{4} = \left(\frac{3}{2} - i\frac{3\sqrt{3}}{2} \right)$	M1 A1 (2)
(b)	$ z_2 = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{3\sqrt{3}}{2}\right)^2} = \sqrt{\frac{9}{4} + \frac{27}{4}}$ The modulus of z_2 is 3 $\tan \theta = (\pm)\sqrt{3}$ and attempts to find θ and the argument is $-\frac{\pi}{3}$	M1 A1 M1 A1 (4)
(c)		M1 A1 (2)
		(8 marks)



(a) M1: Multiplies numerator and denominator by $1-i\sqrt{3}$

A1: any correct equivalent with real denominator.

(b) M1: Uses correct method for modulus for their z_2 in part (a)

A1: for 3 only

M1: Uses tan or inverse tan

A1: $-\frac{\pi}{3}$ accept $\frac{5\pi}{3}$

NB Answers only then award 4/4 but arg must be in terms of π

(c) M1:

Either z_1 on imaginary axis and labelled with z_1 or $3i$ or $(0,3)$ or axis labelled 3;

or their z_2 in the correct quadrant labelled z_2 or $\frac{3}{2}-i\frac{3}{2}\sqrt{3}$ or $\left(\frac{3}{2}, -\frac{3}{2}\sqrt{3}\right)$ or axes labelled

or their $a+bi$ or their (a,b) or axes labelled.

Axes need not be labelled Re and Im.

A1: All 3 correct ie z_1 on positive imaginary axis, z_2 in 4th quadrant and z_1+z_2 in the first quadrant.

Accept points or lines. Arrows not required.

2) (i) The complex number w is given by

$$w = \frac{p-4i}{2-3i}$$

where p is a real constant.

(a) Express w in the form $a+bi$, where a and b are real constants.

Give your answer in its simplest form in terms of p .

(3)

Given that $\arg w = \frac{\pi}{4}$

(b) find the value of p .

(2)

(ii) The complex number z is given by

$$z = (1-\lambda i)(4+3i)$$

where λ is a real constant.

Given that

$$|z| = 45$$

find the possible values of λ

Give your answers as exact values in their simplest form.

(3)

Almost all candidates knew to multiply top and bottom by the complex conjugate of the denominator, and most did this correctly, though there were occasional errors in both numerator and denominator. The final mark in part (i)(a) was sometimes lost due to failure to separate the real and imaginary parts, as the question required. In part (b), relatively few candidates were able to immediately equate the numerators of the real and imaginary parts. Most used $\tan \pi/4 = 1$ involving them in extra work. In part (ii) it was very rare to see use of the modulus of the product as the product of the moduli. Instead candidates worked out the product of the two complex numbers, sometimes incorrectly, and then attempted to obtain the modulus of that product, usually successfully. Most were then able to progress to a correct conclusion, though a worrying minority thought that $\sqrt{(25+\lambda^2)} = (5+\lambda)$.



Question Number	Scheme	Marks
(i)	Mark (i)(a) and (i)(b) together.	
(a) Way 1	$w = \frac{p-4i}{2-3i} \quad \arg w = \frac{\pi}{4}$ $w = \frac{(p-4i)}{(2-3i)} \times \frac{(2+3i)}{(2+3i)}$ $= \left(\frac{2p+12}{13} \right) + \left(\frac{3p-8}{13} \right)i$	M1 Multiplies by $\frac{(2+3i)}{(2+3i)}$ A1 At least one of either the real or imaginary part of w is correct. Must be expanded but could be unsimplified e.g. expressed as single fraction. Condone $a+ib$. A1 Correct w in its simplest form.
(a) Way 2	$(a+ib)(2-3i) = (p-4i)$ $2a+3b = p$ $3a-2b = 4$ $= \left(\frac{2p+12}{13} \right) + \left(\frac{3p-8}{13} \right)i$	M1 Multiplies out to obtain 2 equations in two unknowns. A1 At least one of either the real or imaginary part of w is correct. Must be expanded but could be unsimplified e.g. expressed as single fraction. Condone $a+ib$. A1 Correct w in its simplest form.
(b)	$\left\{ \arg w = \frac{\pi}{4} \Rightarrow \right\} \quad 2p+12 = 3p-8 \text{ o.e. seen anywhere.}$ $\Rightarrow p = 20$	M1 Sets the numerators of the real part of their w equal to the imaginary part of their w or if arctan used, require evidence of $\tan \frac{\pi}{4} = 1$ A1 $p = 20$
(ii) Way 1	$z = (1-\lambda i)(4+3i) \text{ and } z = 45$ $\sqrt{1+\lambda^2} \sqrt{4^2+3^2}$ $\sqrt{1+\lambda^2} \sqrt{4^2+3^2} = 45$ $\{\lambda^2 = 9^2 - 1 \Rightarrow \} \quad \lambda = \pm 4\sqrt{5}$	M1 Attempts to apply $(1-\lambda i)(4+3i) = \sqrt{1+\lambda^2} \sqrt{4^2+3^2}$ A1 Correct equation. A1 $\lambda = \pm 4\sqrt{5}$
Way 2	$z = (4+3\lambda) + (3-4\lambda)i$ $\sqrt{(4+3\lambda)^2 + (3-4\lambda)^2}$ $(4+3\lambda)^2 + (3-4\lambda)^2 = 45^2 \text{ or}$ $\sqrt{(4+3\lambda)^2 + (3-4\lambda)^2} = 45$ $\{16+24\lambda+9\lambda^2+9-24\lambda+16\lambda^2 = 2025\}$ $\{25\lambda^2 = 2000 \Rightarrow \} \quad \lambda = \pm 4\sqrt{5}$	M1 Attempt to multiply out, group real and imaginary parts and apply the modulus. A1 Correct equation. A1 Condone if middle terms in expansions not explicitly stated. A1 $\lambda = \pm 4\sqrt{5}$
Question Notes		
(ii)	M1 Also allow $(1+\lambda^2)(4^2+3^2)$ for M1. M1 Also allow $(4+3\lambda)^2 + (3-4\lambda)^2$ for M1.	

3) The complex number w is given by

$$w = 10 - 5i$$

(a) Find $|w|$.

(1)

(b) Find $\arg w$, giving your answer in radians to 2 decimal places.

(2)

The complex numbers z and w satisfy the equation

$$(2 + i)(z + 3i) = w$$

(c) Use algebra to find z , giving your answer in the form $a + bi$, where a and b are real numbers.

(4)

Given that

$$\arg(\lambda + 9i + w) = \frac{\pi}{4}$$

where λ is a real constant,

(d) find the value of λ .

(2)

Part (a) was usually answered correctly and in part (b) the argument was generally correct, but a minority gave the positive argument by mistake. In part (c) a variety of methods were used to find z . Most candidates were successful, but some had no clear method to account for the two complex numbers when solving. In part (d) many candidates found the value of λ correctly, most of these through the use of trigonometry rather than by equating real and imaginary parts directly.



Question Number	Scheme	EXAM PAPERS PRACTICE	Marks
(a)	$w = 10 - 5i$ $ w = \sqrt{10^2 + (-5)^2} = \sqrt{125} \text{ or } 5\sqrt{5} \text{ or } 11.1803...$	$\sqrt{125} \text{ or } 5\sqrt{5} \text{ or awrt } 11.2$	B1 [1]
(b)	$\arg w = -\tan^{-1}\left(\frac{5}{10}\right)$ $= -0.463647609... = -0.46 \text{ (2 dp)}$	Use of $\tan^{-1}$ or $\tan$ awrt -0.46 or awrt 5.82	M1 A1 oe [2]
(c)	$(2+i)(z+3i) = w$ $z+3i = \frac{10-5i}{(2+i)}$ $z+3i = \frac{(10-5i)}{(2+i)} \times \frac{(2-i)}{(2-i)}$ $z+3i = \frac{20-10i-10i-5}{1+4}$ $z+3i = \frac{15-20i}{5}$ $z+3i = 3-4i$ $z = 3-7i$ (Note: $a=3, b=-7$.)	Simplifies to give $z = \frac{\text{complex no.}}{(2+i)}$ Multiplies by $\frac{\text{their } (2-i)}{\text{their } (2-i)}$ Simplifies realising that a real number is needed on the denominator and applies $i^2 = -1$ on their numerator expression and denominator expression.	B1 M1 M1 A1 [4]
(d)	$\arg(\lambda + 9i + w) = \frac{\pi}{4}$ $\lambda + 9i + w = \lambda + 9i + 10 - 5i = (\lambda + 10) + 4i$ $\arg(\lambda + 9i + w) = \frac{\pi}{4} \Rightarrow \lambda + 10 = 4$ So, $\lambda = -6$	Combines real and imaginary parts and puts "Real part = Imaginary part" i.e. $\frac{\lambda+10}{4} = 1$ or $\frac{4}{\lambda+10} = 1$ o.e.	M1 A1 [2]
(c)	Alt 1: Scheme as above: $(2+i)z + 6i + 3i^2 = 10 - 5i \Rightarrow (2+i)z = 13 - 11i$ B1 for $z = \frac{13-11i}{2+i}$; M1 for $z = \frac{(13-11i)}{(2+i)} \times \frac{(2-i)}{(2-i)}$; M1 for $z = \frac{26-13i-22i-11}{4+1}$; A1 for $z = 3-7i$		
(c)	Alt 2: Let $z = a+ib$ gives $(2+i)(a+ib+3i) = 10-5i$ for B1 Equating real and imaginary parts to form two equations both involving a and b for M1 Solves simultaneous equations as far as $a=$ or $b=$ for M1 $a=3, b=-7$ or $z = 3-7i$ for A1		

4)	$z_1 = 2 + 3i, \quad z_2 = 3 + 2i, \quad z_3 = a + bi, \quad a, b \in \mathbb{R}$	$\frac{z_1 z_3}{z_2}$
(a)	Find the exact value of $ z_1 + z_2 $.	(2)
Given that $w =$,		



(b) find w in terms of a and b , giving your answer in the form $x + iy$, $x, y \in \mathbb{R}$

(4)

Given also that $w = \frac{17}{13} - \frac{7}{13}i$,

(c) find the value of a and the value of b ,

(3)

(d) find $\arg w$, giving your answer in radians to 3 decimal places.

(2)

Part a proved a challenge to some candidates and common errors seen were finding the modulus of z_1 and z_2 , then adding. Unfortunately some candidates just added z_1 and z_2 and made no attempt to find the modulus and a few left i in the square root. Most candidates knew what they were supposed to do to achieve the answer but the amount of simplification required defeated them. Many ended up, after many lines of working, with only one variable in either the real or complex part of their answer. Few failed to spot that multiplying the two numerical terms, $(2 + 3i)$ and $(3 - 2i)$ before multiplying by the algebraic term simplified the working.

In part (c) the algebra involved in achieving an answer, following an earlier error, proved to be a challenge.

The mark for the use of $\tan$ was usually achieved in part (d) and common errors in the final answer were omitting the negative sign or inverting the fraction.



Question Number	Scheme	Notes	Marks
	$z_1 = 2 + 3i, \quad z_2 = 3 + 2i$		
(a)	$z_1 + z_2 = 5 + 5i \Rightarrow z_1 + z_2 = \sqrt{5^2 + 5^2}$	Adds z_1 and z_2 and correct use of Pythagoras. i under square root award M0.	M1
	$\sqrt{50} (= 5\sqrt{2})$		A1 cao
			(2)
(b)	$\frac{z_1 z_3}{z_2} = \frac{(2 + 3i)(a + bi)}{3 + 2i}$	Substitutes for z_1, z_2 and z_3 and multiplies by $\frac{3 - 2i}{3 - 2i}$	M1
	$= \frac{(2 + 3i)(a + bi)(3 - 2i)}{(3 + 2i)(3 - 2i)}$		
	$(3 + 2i)(3 - 2i) = 13$	13 seen.	B1
	$\frac{z_1 z_3}{z_2} = \frac{(12a - 5b) + (5a + 12b)i}{13}$	M1: Obtains a numerator with 2 real and 2 imaginary parts. A1: As stated or $\frac{(12a - 5b)}{13} + \frac{(5a + 12b)}{13}i$ ONLY.	dM1A1
			(4)
(c)	$12a - 5b = 17$ $5a + 12b = -7$	Compares real and imaginary parts to obtain 2 equations which both involve a and b . Condone sign errors only.	M1
	$60a - 25b = 85$ $60a + 144b = -84 \Rightarrow b = -1$	Solves as far as $a =$ or $b =$	dM1
	$a = 1, b = -1$	Both correct	A1
		Correct answers with no working award 3/3.	
			(3)
(d)	$\arg(w) = -\tan^{-1}\left(\frac{7}{17}\right)$	Accept use of $\pm \tan^{-1}$ or $\pm \tan$. awrt ± 0.391 or ± 5.89 implies M1.	M1
	$= \text{awrt } -0.391 \text{ or awrt } 5.89$		A1
			(2)
			Total 11